Discharging H-diag I: the full-covariance Gaussian-Hartree problem,

TECT verification-first repository · claim B2-PROPA-HLAYER

first issued 2026-06-09 · this version issued 2026-06-09 · v1.0

1. Purpose and scope

B1-RH-ENUM rests on the sole hypothesis H-LAYER, whose analytic core is RES-1: Math427 proves the isotropic dressing is the variational infimum only within the DIAGONAL Gaussian trial class (conditional on H-diag = diagonal covariance + local Wick); the off-diagonal (condensate-induced Bogoliubov-band, G1″) sector is the live gate. This note begins the discharge of H-diag by (i) writing the variational problem over the FULL covariance operator, (ii) proving rigorously that the functional is convex absent the condensate-induced off-diagonal coupling, and hence (iii) isolating the entire obstruction to that one coupling. No discharge is claimed: step (iii) leaves a named open gap. Scope: second-cumulant (Gaussian-Hartree) order, the production anchor $\mu^2 = 5 \times 10^{-3}$.

2. The full-covariance variational functional

For a homogeneous Gaussian trial state with covariance operator G (a positive operator on momentum space, $G(q, q')$; the diagonal class is $G(q, q') = G(q)\delta_{qq'}$),

$$\frac{F[G]}{V} = \underbrace{\tfrac{1}{2}\,\mathrm{Tr}\,\ln G^{-1}}_{\text{entropy}} + \underbrace{\tfrac{1}{2}\,\mathrm{Tr}\,[K_0\,G]}_{\text{kinetic (linear)}} + \Phi[G], \qquad K_0(q) = c\,(q^2 - q_0^2)^2,$$

where $\Phi[G]$ is the interaction (quartic u, sextic v, condensate A). In the DIAGONAL class the local quartic/sextic interactions make Φ depend on the trial only through the scalar total variance $M_{\mathrm{tot}} = \int_q G(q)$ (Math427), giving $\Phi_{\mathrm{diag}}(M) = \frac{3u}{4}M^2 + \frac{5v}{2}M^3$ after Wick reduction, and the stationarity $\hat{r}(\hat{q}) = 2\Phi'(M_{\mathrm{tot}})$ is direction-independent $\Rightarrow$ isotropic. Off the diagonal, the condensate A (with momentum content on the Bragg shell $|k| = q_0$) couples $G(q, q')$ for $q' - q$ a reciprocal-lattice vector – a periodic-potential / Bogoliubov-band term that breaks the "Φ depends only on M_{tot}" structure. Call this term $\Phi_{\mathrm{od}}[G]$.

3. The condensate-free functional is convex (rigorous)

Lemma (condensate-free convexity). The functional $F_0[G] = \frac{1}{2}\mathrm{Tr}\ln G^{-1} + \frac{1}{2}\mathrm{Tr}[K_0 G] + \Phi_{\mathrm{diag}}(M_{\mathrm{tot}})$ (i.e. $F[G]$ with Φ_{od} omitted) is convex on the positive-covariance cone at the operating point, and its unique minimiser is the isotropic dressing.

Proof. (a) $\frac{1}{2}\mathrm{Tr}\ln G^{-1} = -\frac{1}{2}\mathrm{Tr}\ln G$ is convex in G ($-\ln\det$ is operator-convex; per-mode Hessian $\frac{1}{2}G^{-2} > 0$, verified $\geq 4.6 \times 10^{-2}$ on the BZ grid). (b) $\frac{1}{2}\mathrm{Tr}[K_0 G]$ is linear in G. (c) $\Phi_{\mathrm{diag}}(M_{\mathrm{tot}})$ is a convex function of the linear functional $M_{\mathrm{tot}}[G]$: $\Phi''_{\mathrm{diag}}(M) = \frac{3u}{2} + 15vM = \frac{3}{2}(u + 10vM) = \frac{3}{2}u_{\mathrm{eff}}$, and at the anchor $u_{\rm eff} = u + 10vM_R = -0.86 + 10(3.24)(0.10941) = +2.685 > 0$, so $\Phi''_{\rm diag} = +4.03 > 0$ and (with $v = 3.24 > 0$) the sextic gives boundary coercivity $(5v/2)M^3 \to +\infty$. A sum of convex + linear + (convex of linear) is convex; coercive ⇒ the stationary isotropic point is the unique global minimiser. □

This UPGRADES Math427's diagonal "infimum" (which invoked a gap-map monotonicity) to a convex-unique GLOBAL minimum, and – crucially – it holds over the FULL covariance cone, not only the diagonal class: F_0 is convex in every off-diagonal direction as well. Hence the diagonal restriction H-diag is unnecessary for F_0.

Honest subtlety. The bare quartic is ATTRACTIVE ($u = -0.86 < 0$, the Brazovskii sign); convexity is NOT automatic. It holds because the sextic dressing lifts the effective quartic to $u_{\rm eff} = +2.685 > 0$ at the anchor. Off the anchor, $u_{\rm eff}$ could in principle change sign – this is exactly the ROBUSTNESS-MU2 axis (closed $[\times 0.5, \times 2]$, where $u_{\rm eff}$ stays positive), and the convexity statement is anchored accordingly.

4. The isolated obstruction (the RES-1 residual)

Since $F = F_0 + \Phi_{\rm od}$ and F_0 is convex with the isotropic global minimum, the ONLY possible source of a lower off-diagonal competitor is the condensate-induced term $\Phi_{\rm od}$. Discharging H-diag is therefore EQUIVALENT to the Bogoliubov-stability inequality

$$\boxed{\mathrm{Hess}\, F\big|_{G_*} \succeq 0 \iff \underbrace{\tfrac{1}{2} G_*^{-1} \otimes G_*^{-1}}_{\text{entropy curvature (PD)}} \succeq -\,\mathrm{Hess}\,\Phi_{\rm od}\big|_{G_*}}$$

at the isotropic dressing G_* (and, for a GLOBAL statement, convexity of $\Phi_{\rm od}$ or a global enclosure). The entropy curvature is positive-definite and large where G_* is small (deep in the gap); the question is whether the condensate-induced off-diagonal kernel – which can open attractive band gaps even for repulsive $u_{\rm eff}$ – is dominated by it. For the five ENUMERATED readings this was executed (Math428, the G1$''$ Bogoliubov bands); the CLASS-WIDE positivity is the open RES-1 content, and its beyond-second-cumulant part merges with RES-5.

5. Status and next step

T3 PROOF SKETCH. Proved: F_0 convex (isotropic global min over the full covariance cone) at the anchor. Formulated: H-diag discharge $\iff$ the condensate-induced off-diagonal Hessian $\mathrm{Hess}\,\Phi_{\rm od}$ is dominated by the entropy curvature. **OPEN GAP (RES-1):** the class-wide positivity of $\mathrm{Hess}\,\Phi_{\rm od}|_{G_*}$ (second cumulant), and its beyond-second-order extension (RES-5). No tier flip: B2-PROPA-HLAYER T6, B1-RH-ENUM T6 on {H-LAYER}.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"The bare quartic is attractive ($u < 0$); the convexity claim is fragile."* VALID-with-mitigation: stated explicitly in §3 – convexity rests on $u_{\rm eff} = +2.685 > 0$ from the sextic dressing at the anchor; the off-anchor sign is the ROBUSTNESS-MU2 axis (closed $[\times 0.5, \times 2]$). The Lemma is anchored, not claimed universally in μ^2.

(β) *"Isolating the obstruction is not discharging H-diag."* DISMISSED (by design): the note claims T3, proving only the condensate-free convexity and reducing the discharge to the boxed Bogoliubov inequality; the off-diagonal positivity is the marked OPEN GAP, not a result.

(γ) *"A condensate-induced periodic potential opens band gaps even for repulsive $u_{\rm eff}$, so $\Phi_{\rm od}$ could lower F."* VALID and UNRESOLVED: this is precisely the RES-1 residual. The note does NOT assert $\operatorname{Hess}\Phi_{\rm od} \succeq 0$ class-wide; it only localises the entire risk to this one term, which is the value of the reduction.

Quantitative sanity check (mandatory): *sign* – bare $u = -0.86 < 0$ (attractive) but $u_{\rm eff} = u + 10vM_R = +2.685 > 0$ (the dressing flips the sign), so $\Phi''_{\rm diag} = \frac{3}{2}u_{\rm eff} = +4.03 > 0$; *magnitude* – the entropy Hessian $\frac{1}{2}G^{-2} \geq 4.6 \times 10^{-2}$ on the BZ grid (positive-definite); *limit* – $(5v/2)M^3 \to +\infty$ as $M \to \infty$ ($v = 3.24 > 0$), coercive; *consistency* – the convex global diagonal minimum reproduces the Math427 isotropic stationary point.

7. Result footer

```
Result ID:              B2-PROPA-HLAYER / H-diag discharge I (RES-1 formulation +
                        condensate-free convexity)
Precise statement:      (1) F0[G] = entropy + linear kinetic + Phi_diag(M_tot),
                        with Phi_diag''=(3/2)u_eff=+4.03>0 (u_eff=+2.685 from the
                        sextic dressing of the attractive bare u=-0.86) and the
                        entropy Hessian (1/2)G^-2>0, is CONVEX on the full
                        covariance cone at the anchor => the isotropic dressing
                        is its unique GLOBAL minimum (diagonal AND off-diagonal).
                        (2) Therefore H-diag discharge <=> the condensate-induced
                        off-diagonal Hessian Hess Phi_od|_G* is dominated by the
                        entropy curvature (1/2)G*^-1 (x) G*^-1 (Bogoliubov
                        stability). No discharge claimed.
Scope:                  Second-cumulant (Gaussian-Hartree) order, anchor mu^2;
                        the condensate-free convexity is rigorous, the
                        off-diagonal positivity is OPEN.
Dependencies:           Math427 (diagonal isotropy infimum); Math424-AddA
                        (gap_solve, M_fast, U=-0.86, V=+3.24); Math428 (G1''
                        enumerated); hlayer-residual-inventory (RES-1/RES-5).
Evidence grade:         ANALYTIC (convexity) + EXECUTED (hdiag_convexity_probe.py
                        5/5).
Reproduction command:   python codes/vacuum/hdiag_convexity_probe.py
Expected output:        u_eff=+2.685>0; Phi''=+4.03>0; entropy Hessian>0; 5/5.
Falsification gate:     u_eff<=0 at the anchor (would void the convexity), or a
                        demonstrated off-diagonal competitor with F<F[G*] (would
                        show Hess Phi_od is NOT dominated -- the residual decided
                        negatively).
Tier before / after:    No tier flip. B2 T6, B1 T6 on {H-LAYER} unchanged.
                        H-diag: restated as the boxed Bogoliubov inequality with
                        the condensate-free half PROVED (T3 proof sketch).
No-overclaim statement: This does NOT discharge H-diag or H-LAYER and does NOT
                        promote B1. Only the condensate-FREE convexity is proved;
                        the condensate-induced off-diagonal (G1'') Hessian
                        positivity is the OPEN RES-1 gap (class-wide), merging
                        with RES-5 beyond second cumulant. Convexity is anchored
                        (bare quartic attractive; u_eff>0 via dressing).
Next required action:   Bound/verify Hess Phi_od|_G* class-wide (the condensate
                        off-diagonal Bogoliubov Hessian): construct it from the
                        Bragg-shell condensate coupling, test PSD against the
                        entropy curvature for a general single-mode pattern.
```